```
cout<<"\n The sum of 1 to " <<n<<"is "<<sum;
return 0;
}
```

Output

```
Enter the value of n 5
The sum of 1 to 5 is 15
```

In the above example, the variable i is declared and sum is initialized at the time of variable declaration. The variable i is assigned to 0 before the for loop but still the semicolon is necessary before test expression. In a for loop, if the initialization expression is absent then the control is transferred to test expression/conditional part.

Case 2

Illustration 10.13 (b) C++ program to sum the numbers from 1 to 10

```
#include <iostream>
using namespace std;
int main ()
{
int i, sum=0, n;
cout<<"\n Enter The value of n";
cin>>n;
i =1;
for (  ; i<=10;  )
 {
        sum += i;
        ++i;
 }
cout<<"\n The sum of 1 to " <<n<<"is "<<sum;
return 0;
}
```

Output

```
Enter the value of n 5
The sum of 1 to 5 is 15
```